



Pure substances

Task A: Pure substances

Look carefully at the picture.



“pure orange juice”

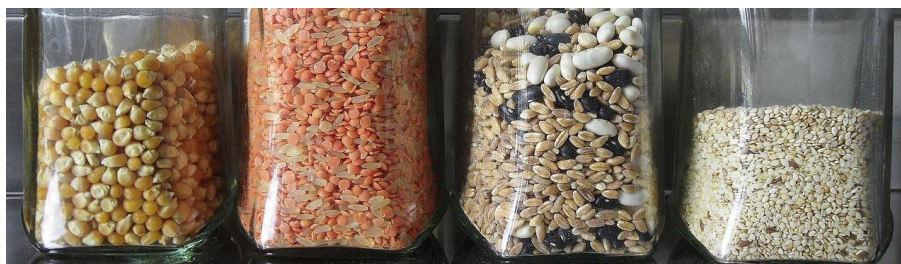
- 1) Is the term **pure** being used scientifically?
*No, **pure** is not being used scientifically.*

- 2) Justify your answer.

An orange contains several chemicals. For example: carbohydrates, proteins, vitamin C, and potassium.

Task B: Mixtures

Use the words **pure**, **mixture**, or **impurity** to write a sentence describing each model shown.



a

b

c

d

Your answers may be similar to these:

- Jar A is a model of a **pure** material because it only contains one grain - popcorn kernels.
- Jar B is a model of a **mixture** because it contains two grains - lentils and rice - which can be separated.
- Jar C is a model of a **mixture** because it contains three substances - brown rice, black beans and white beans - which can be separated.
- Jar D is a model of a **mixture** because it contains two substances which can be separated; however, it appears to have an **impurity** of a few darker grains.



Task C: How to identify pure substances and mix

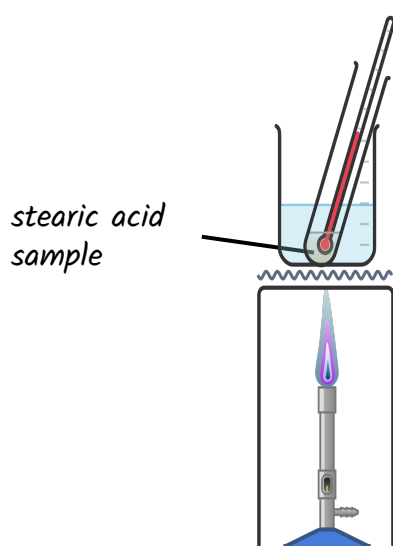
A lab technician has two jars of stearic acid, which is a waxy solid at room temperature.

She suspects the stearic acid in one of the jars is contaminated with an **impurity**.



- 1) Describe an experiment you could perform to decide if one of the jars is contaminated.
She can let you have 2 g from each jar.

Heat the stearic acid sample until it melts and measure the temperature (or temperature range) at which this happens.



- 2) What result would you expect if the sample is **pure**?

*The **pure** sample will melt at a specific temperature.*

- 3) What result would you expect for a contaminated sample?

*The contaminated sample will melt over a range of temperatures, because it is a **mixture** - stearic acid and an **impurity**.*

